The Hong Kong University of Science and Technology
Department of Electronic and Computer Engineering

ELEC2400 **ELECTRONIC CIRCUITS** FALL 2026-27

Lab Section No. : _____ Date : _____

Name : _____ Student Number : _____

Name of the group partner : _____

LAB 1(b) – Instrumentation and Measurement

Answer Sheet

Q7. Sketch the waveform of the triangle wave, with clear labels. (Screen capture is not accepted)

Q8. What is the frequency of the triangle wave measure in the scope? Frequency = _____

Q9. Sketch the waveform of the resultant waveform '**A + B**', with clear labels. (Screen capture is not accepted)

Q10. What is the frequency of the resultant waveform? Frequency = _____

Q11. What is $\mathbf{V_{DMM}}$ (square)? _____

Q12. What is $\mathbf{V_{DMM}}$ (triangular)? _____

Q13. What is $\mathbf{V_{DMM}}$ (sine)? _____

Q14. What is theoretical RMS value of **V**(square)? _____

Q15. What is theoretical RMS value of **V**(triangular)? _____

Q16. What is theoretical RMS value of **V**(sine)? _____

Q17. What is V_C (DMM)? _____

Q18. What is V_A (DMM)? _____

Q19. What is V_B (DMM)? _____

Q20. What is V_A (x1)? _____

Q21. What is V_B (x1)? _____

Q22. What is V_A (x10)? _____

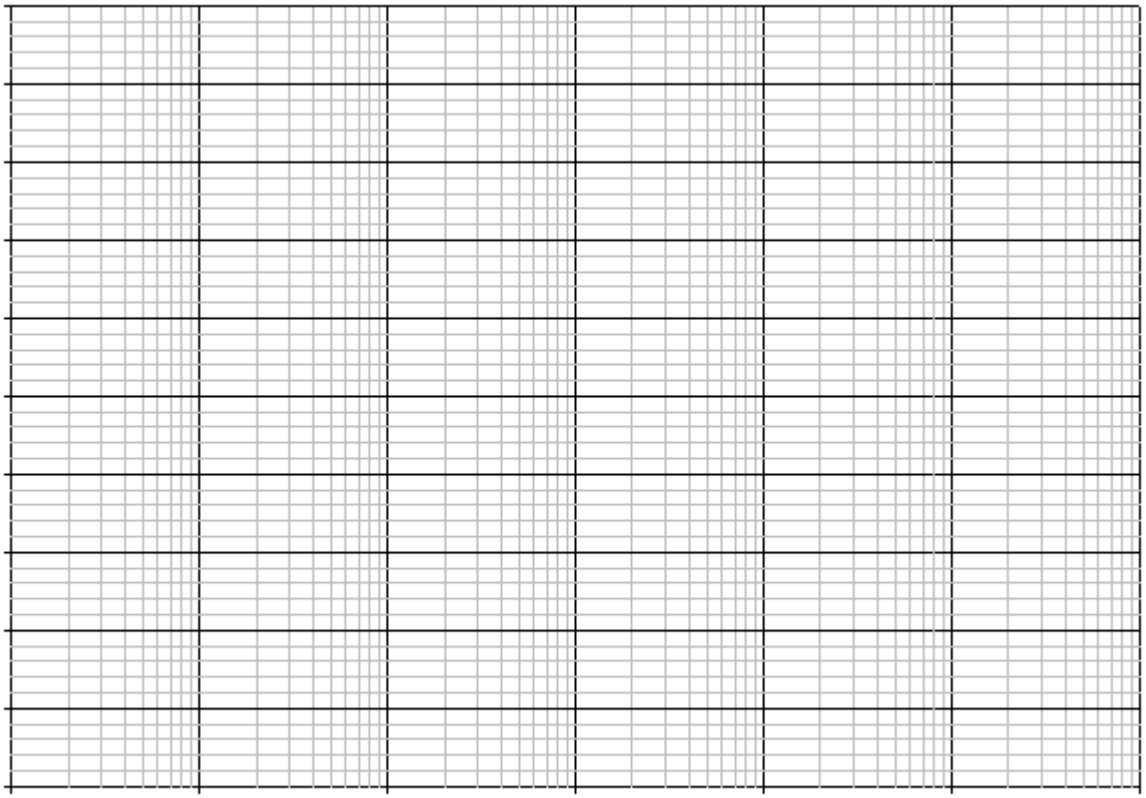
Q23. What is V_B (x10)? _____

Q24. Explain briefly the results in 5.4.2., 5.4.3 and 5.4.4.

Q25. Fill in the data in the following table

Frequency (Hz)	Voltage output (V)
100	
500	
1000	
1200	
1400	
1600	
1800	
2000	
2200	
2400	
2600	
2800	
3000	
3500	
4000	
5000	
6000	
7000	
8000	
9000	
10000	
15000	
20000	

Q26. Sketch the V_{DMM} versus frequency results obtained in sections 5.5.2 and 5.5.3. Use a logarithmic scale for the frequency on the semi-logarithmic graph paper. (Do not plot it using computer software)



Q27. What is the estimated bandwidth of the DMM from the plot in Q26? (show how to estimate the bandwidth from the plot in Q26) _____

Q28. Show the circuits and waveform to TA.

TA (Demonstration code) : _____